

China Business Trip Report

PTC Visual Machine BB087

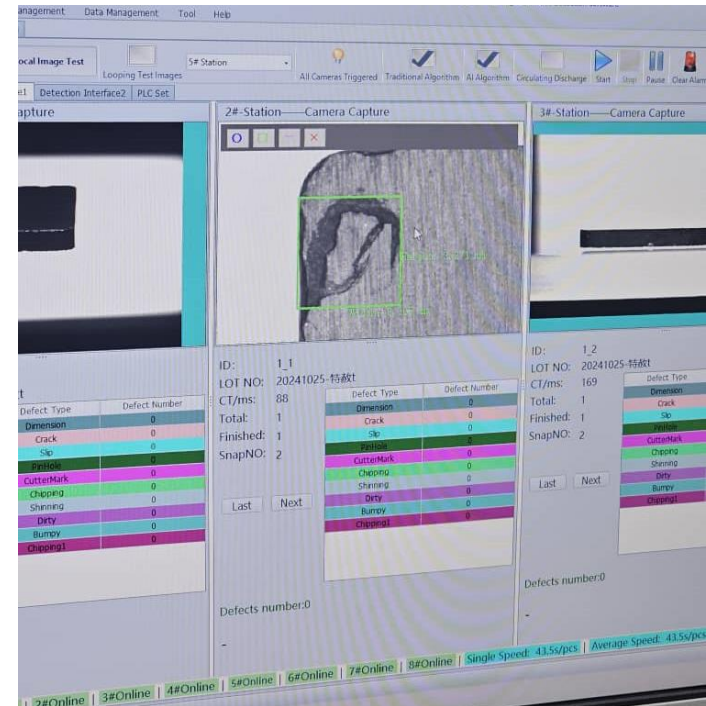
(PTC AOI)

Date : 25/10 (DAY 5)

Meeting member : Mr Afiq (Maruwa)
Mr Kitauji (Maruwa)
Mr Miyano (Maruwa Japan)

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Time : 10.00 am – 5.00 pm



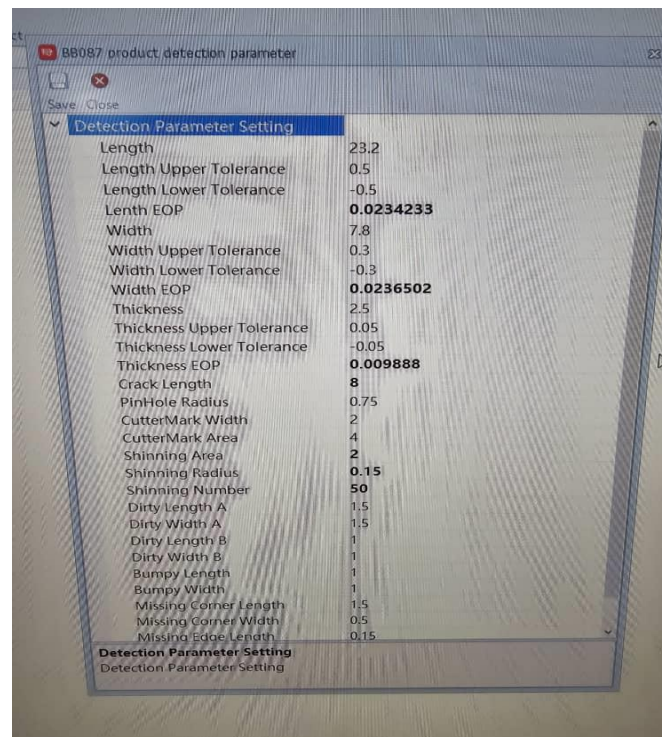
- Day 5 consist on consensus agreeable between two PTC and Maruwa regarding limit NG of each defects. Including improvement to the machine regarding machine/operator safety during AOI operation.
- Include AI improvement adjustment to the BB087 measurement, side chipping limit, crack limit, pinhole limit, slip limit and other defects.

AOI/Operator Safety



➔ PTC will include additional emergency button at output for AOI machine/ operator safety during emergency.

Defects Parameter Lock



➔ PTC will include pass lock for NG/OK limit parameter.
 (Only Authorize Staff can adjust the NG limit)

→ The NG limit consensus is discuss between PTC and Maruwa on Day 5



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Magnet Surface Inspection Equipment Consensus Content

Data: October 21, 2024 - October 25, 2024

Participants: Both teams

■ Consensus content

1. Pinhole judging criteria: diameter $> 1.5\text{mm}$ judged as NG;
2. Slip judgment criteria: $1/3$ of defect length $>$ total length around the product surface is judged as NG;
3. The judging criteria for the front and back of Chipping are as follows (the four perimeters are defined as Length, and the vertical direction with the edge is defined as Width) ;



The calculable area of the chipping defect in the middle of the product is detected as the Pinhole defect;

- ① $L \leq 4\text{mm}$, $W \leq 3\text{mm}$, judged as OK; if $L \leq 4\text{mm}$ and $W > 3\text{mm}$, it is judged as NG;
- ② $4\text{mm} < L \leq 5\text{mm}$, $W \leq 1.5\text{mm}$, judged OK; if $4\text{mm} < L \leq 5\text{mm}$, $W > 1.5\text{mm}$, judged as NG;
- ③ $L > 5\text{mm}$, $W \leq 0.5\text{mm}$ judged as OK; if $L > 5\text{mm}$, $W > 0.5\text{mm}$ as NG;
- ④ The calculable area of the chipping defect in the middle of the product is detected as the Pinhole defect;

➔ The NG limit consensus is discuss between PTC and Maruwa on Day 5



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4. The judging criteria for Chipping are as follows:



In the blue area (1.5mm) : chipping is OK regardless of size;

In the green area:

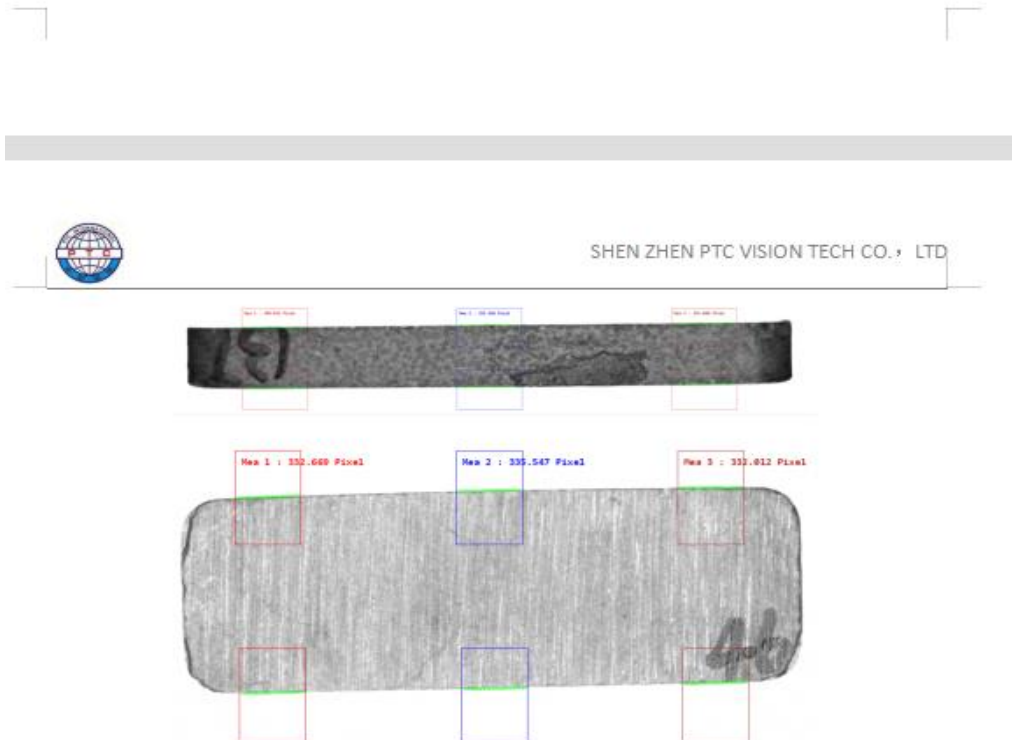
- ①chipping high $H < 0.5\text{mm}$ is OK;
- ②If the chipping height is $0.5\text{mm} < H < 1\text{mm}$ and $L < 4\text{mm}$, it is Ok;
- ③The chipping height $H > 1\text{mm}$ is NG;

As shown in the picture:

- 5. Judging criteria of Crack: defect diagonal length $> 8\text{mm}$ is judged as NG;
- 6. Judging criteria for Bumpy: Defect length or width $> 1\text{mm}$ is judged as NG;
- 7. Dirty judging criteria: front and back length or width $> 1.5\text{mm}$, side length or width $> 1\text{mm}$ judged as NG;
- 8. Shinning criteria: According to the current criteria (shinning Area= 1.2mm^2 , shinning radius= 0.1mm , shinning number=30) ;
- 9. Cutter Mark judging criteria: according to the current judging criteria (Cutter Mark Width= 2mm , Cutter Mark Area= 4mm^2);

➔ The NG limit consensus is discuss between PTC and Maruwa on Day 5

10. Dimension judgment criteria: 3 points on the long side, 2 points on the short side,
and 3 points on the height;



➔ The network IP is adjusted for IOT integration (Maruwa IP Network)

11. Change the IP address as follows:

The IP address used by the equipment is shown in the table (provided by the customer) :

Hi Afiq
For the FM Auto Visual machine, kindly inform to the supplier to revise the current IP setting to new setting as per you need additional IPs

	Current Setting	Change to
电脑IP1	192.168.4.190	192.168.4.183
电脑IP2	192.168.4.191	192.168.4.184
PLC IP	192.168.4.192	192.168.4.185
视频监控IP	192.168.4.193	192.168.4.186
光源控制器IP1	192.168.4.194	192.168.4.187
光源控制器IP2	192.168.4.195	192.168.4.188
光源控制器IP3	192.168.4.196	192.168.4.189

Regards
Siva



Conclusion Day 5:

➔ PTC currently ongoing AOI machine improvement based on requirement requested by Maruwa **(Schedule completion on 31/10).**



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■ Schedule

No	Specific item	Completion Time	State
1	Shoot video of equipment debugging operation;	2024/10/29	Open
2	Dimension software data statistics: 3 values are presented for height, 3 values for width, and 2 values for length;	2024/10/30	Open
3	The part of the vibration disc is connected to the route bundle of the detection machine by the aviation joint, which is not fixed on the hood panel;	2024/10/30	Open
4	An emergency stop button is added to the discharging direction device;	2024/10/30	Open
5	The main power plug of the equipment is changed to the British plug;	2024/10/30	Open
6	Machine alarm log can be exported separately;	/	Closed
7	Chipping parameter is open;	/	Closed
8	The vibration disk control part must be labeled in English.	/	Closed
9	On the software interface, the STOP and PAUSE ICONS are exchanged.	/	Closed
10	The identifiers of light source and camera inside the machine have been replaced with English identifiers;	/	Closed
11	Slip parameter is open;	/	Closed
12	Provide electrical material list; (Attached: Brand model/drawing)	2024/10/30	Open
13	Provide a list of spare parts; (Attached: Brand model/drawing)	2024/10/30	Open
14	Add whether the defect category needs to be checked;	2024/10/31	Open
15	Mark each screw before shipment;	2024/10/30	Open
16	Instructions for replacing the glass turntable (steps + precautions);	2024/10/30	Open
17	Set algorithm parameters interface to add engineer permission password;	2024/10/30	Open
18	Provide PLC ladder diagram annotation document;	2024/10/28	Open

➔ AOI Machine will be deliver to Maruwa Malaysia after all necessary improvement regarding AOI mechanical and NG AI detection software complete adjustment.



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